Magic Gems: A Polyhedral Framework for Magic Squares

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Abstract

A magic square can be read as an arrangement of vectors from the centroid whose vertical components cancel along every row, column and both diagonals. The *Magic Gem* of an $n \times n$ magic square is the convex hull of the tips $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij})$ of these vectors, where (x_{ij}, y_{ij}) is the centred cell and z_{ij} the centred entry. We show that $\sum \mathbf{p}_{ij} = \mathbf{0}$ for every arrangement, so unweighted vector balance is not specific to magic squares; the magic-specific consequences are the height-weighted moments $\sum z\,\varphi(x) = \sum z\,\varphi(y) = 0$, which include Loly's invariant, while $\sum z\,xy$ is not forced. For order four we compute the 880 gems exactly (8 to 16 vertices, 39 f-vectors, 352 combinatorial types, 155 volumes, 616 congruence classes) and, by an exact computer-assisted calculation, determine all their isometry groups: the 48 associative squares give centrally symmetric gems, the 304 remaining self-complementary squares give gems with exactly one half-turn symmetry, and the other 528 gems have none; every gem is chiral except the 48 centrally symmetric ones. Every gem lies in a box that depends only on n, certain corner and side-extreme points are always vertices, and the 880 gems have 554 distinct Ehrhart polynomials. We also prove, with computer assistance, that the Siamese square of every odd order $n \geq 5$ has a gem with exactly twelve vertices.

Keywords: magic squares, convex polyhedra, symmetry, combinatorial geometry, moment of inertia

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu belongs to Chinese tradition, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order have been known for centuries [2, 18]. The modern literature treats them from several directions: enumeration and the geometry of the polytopes whose lattice points are the squares [22, 9, 1, 5, 4], spectral and matrix properties [14, 16, 17, 6], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [12, 19, 20].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \ldots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $M(n) = n(n^2 + 1)/2$, and *semi-magic* if only the rows and columns do.

The picture. Centre the grid at the origin and attach to the cell (i, j) the vector

$$\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}),$$

whose horizontal part (x_{ij}, y_{ij}) is the position of the cell and whose vertical part is the *centred entry* $z_{ij} = s_{ij} - \frac{n^2+1}{2}$. Then S is magic exactly when the vertical components of these vectors cancel along every row, column and both diagonals (Lemma 2.2; Figure 1, Figure 2). The *Magic Gem* $\mathcal{G}(S)$ is the convex hull of the tips of the n^2 vectors $\mathbf{p}_{ij}$ (Section 2). Figure 1 shows the vectors of

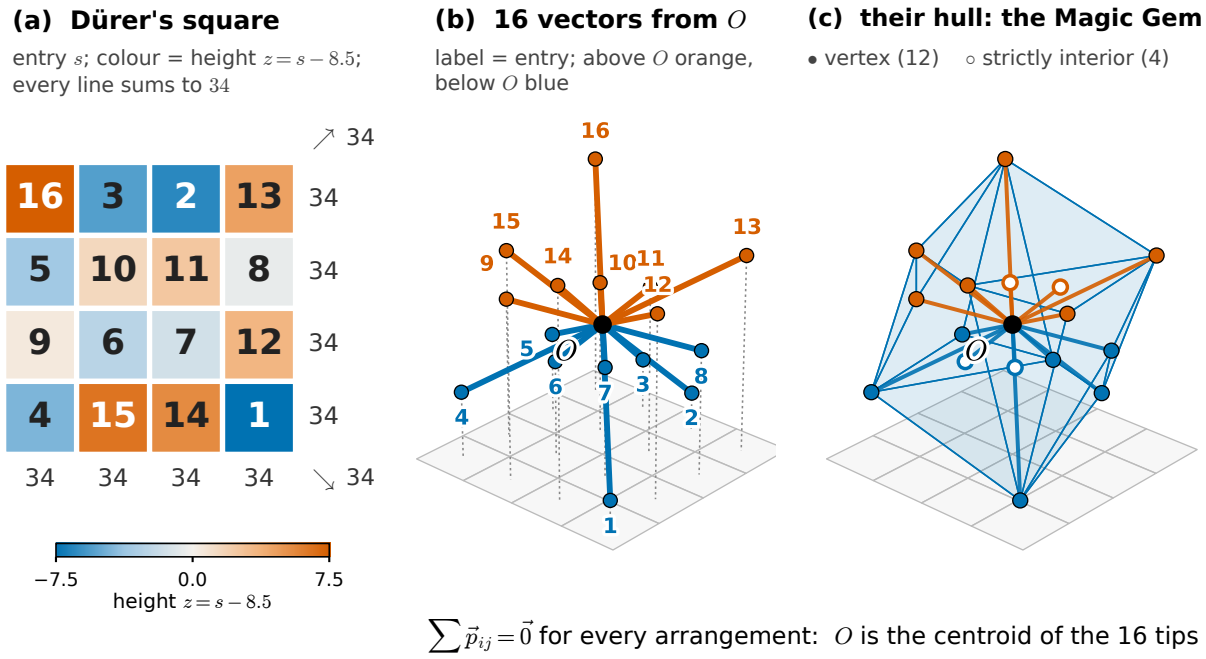


Figure 1: Dürer's square as vectors from the centroid O. (a) The square as a heat map of the heights $z_{ij} = s_{ij} - 8.5$ (orange: above O, blue: below), with the entries s_{ij} printed in the cells; every row, column and both diagonals sum to 34. (b) The 16 vectors $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij})$: each points from O over the cell (x_{ij}, y_{ij}) of the floor grid (dotted drop lines) to a tip at height z_{ij}, labelled by the entry s_{ij}. (c) The same vectors inside their convex hull, the Magic Gem: 12 tips are vertices (filled dots) and 4 are strictly interior (open dots). The colour bar is shared by (a) and (b). The vectors sum to $\mathbf{0}$ for every arrangement of $1, \ldots, 16$ (Proposition 3.1); what singles out magic squares is how the heights cancel along lines (Figure 2).

Dürer's square and the gem they span; the tips are the gem's 12 vertices and four strictly interior points, and $\sum \mathbf{p}_{ij} = \mathbf{0}$.

An interactive display of the 880 order-four gems is available at `https://kylemath.github.io/MagicGemWebpage/`.

Results.

1. *Balance* (Section 3). For every arrangement, magic or not, $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$ (Proposition 3.1), so the unweighted statement "the vectors balance about the centroid" carries no information about S; it concerns all n^2 points, including those that are not vertices, not the solid hull. The magic-specific statements weight each vector by its height: $\sum z\,\varphi(x) = \sum z\,\varphi(y) = 0$ for every function φ when S is semi-magic, which contains the vanishing of the height-weighted centroid and Loly's invariant, while the mixed moment $\sum z\,xy$ is not forced (Proposition 3.2, Corollary 3.3).

2. *Symmetry of the order-four gems* (Section 4). The complement $S \mapsto 17 - S$ reflects the gem in the plane $z = 0$. We prove that a self-complementary square yields a gem with an explicit isometry (a point reflection if S is associative, a half-turn about a horizontal axis otherwise), and determine by exact computation the complete isometry group of each of the 880 gems: 48 gems are centrally symmetric, 304 have exactly one half-turn symmetry, and 528 have no symmetry (Theorem 4.3). Facet orbits are classified (Proposition 4.4).

3. *Geometry of the* 880 *gems* (Section 5.1). Vertex counts, f-vectors, combinatorial types, volumes and congruence classes; the congruence classes are exactly the orbits of the complement map, and no two gems have the same vertex set. Every gem lies in a fixed box, and the points over the corners and the extreme entries of each side are always vertices (Proposition 5.2); the order-four gems have 554 distinct Ehrhart polynomials (Section 5.3).

4. *Siamese gems* (Proposition 5.3). The Siamese square of every odd order $n \geq 5$ has a gem with exactly twelve vertices, 26 edges and 16 facets (computer-assisted proof, Section B); the fraction of points that are vertices tends to zero.

What is and is not new. The polyhedral geometry of magic squares is classical: nonnegative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope, and their number is a polynomial in t [22, 9, 25]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [5, 4]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; contingency tables with fixed margins are the same lattice points seen as a sample space [7]. In all of these the magic squares are the *points* of a polytope, whereas the Magic Gem attaches a polyhedron to a *single* square, in the spirit of the physical invariants of Loly and Rogers [12, 19, 20] (see also [13] for Franklin squares). The balance statements of Section 3 are elementary and implicit in that work; we state and prove them in the gem's coordinates because the geometric picture invites misreadings of them. The symmetry classification, the exact geometry of the 880 gems (including the forced-vertex and Ehrhart statements) and the Siamese theorem are new. The reading of the line conditions as orthogonality conditions on the height function, the hierarchy of moment conditions, the statistics of random arrangements and the enumeration of low-energy squares are developed in the companion paper [15]; none of its results is used here.

Section 2 defines the gem, Section 3 treats balance, Section 4 symmetry, Section 5 the exact geometry of the order-four and Siamese gems, Section 6 open questions, and the appendices contain the computational methods and the remaining steps of the Siamese proof.

2 The Magic Gem

Let $S = (s_{ij})$ be an arrangement of order n, with cells indexed by $i, j \in \{0, \ldots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$.

Definition 2.1 (Gem coordinates and Magic Gem)**.** To the cell (i, j) assign the point $\mathbf{p}_{ij} = (x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ with

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}. \tag{1}$$

The *Magic Gem* of S is the convex hull $\mathcal{G}(S) = \operatorname{conv}\{\mathbf{p}_{ij}\}$. The function $z = (z_{ij})$ is the *height function* of S, and each $\mathbf{p}_{ij}$ is read as a vector from the origin. Every centroid or moment statement in this paper refers to the full set of N points $\mathbf{p}_{ij}$, including those that are not vertices, and not to the solid $\mathcal{G}(S)$.

Terminology. Every point $\mathbf{p}_{ij}$ is of exactly one of three kinds: a *vertex* (an extreme point of $\mathcal{G}(S)$); a *strictly interior point* (lying in the topological interior of $\mathcal{G}(S)$); or a *boundary non-vertex point* (lying on the boundary, in the relative interior of an edge or of a facet). If V, I, B denote their numbers, $V + I + B = N$ and the number of hull vertices is V; the f-vector is (V, E, F).

The x-axis runs along the columns and the y-axis along the rows (top row at positive y); the centred entries have mean zero because the mean entry is $(N+1)/2$. Let $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ denote the n values taken by x and y. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n - 1\}$; each contains n cells. For a line L let $\delta_L(S) = \sum_{c \in L} s_c - M(n)$ be its *deviation*, so S is magic iff $\delta_L(S) = 0$ for all $2n + 2$ lines.

Lemma 2.2 (Means, variances, line vector sums)**.** *For every arrangement S of order n:*

1. $\sum_{i,j} x_{ij} = \sum_{i,j} y_{ij} = \sum_{i,j} z_{ij} = 0$ *and* $\sum_{i,j} x_{ij} y_{ij} = 0$*;*
2. *the empirical variances are* $\operatorname{Var}(x) = \operatorname{Var}(y) = \frac{n^2-1}{12}$ *and* $\operatorname{Var}(z) = \frac{n^4-1}{12}$*;*

3. for every line L, $\delta_L(S) = \sum_{c\in L} z_c$, and

$$\sum_{c\in L} \mathbf{p}_c = \big(n\,\bar{x}_L,\ n\,\bar{y}_L,\ \delta_L(S)\big),$$

where $(\bar{x}_L, \bar{y}_L)$ is the centre of the line (the mean of its cell positions): $(0, y_i)$ for the row i, $(x_j, 0)$ for the column j, and $(0,0)$ for both diagonals.

In particular S is magic if and only if the vector sum of the $\mathbf{p}_c$ along every line is horizontal, i.e. the vertical components cancel along every row, column and diagonal.

Proof. (1) $\sum_{i,j} x_{ij} = n\sum_j x_j = 0$, likewise for y; $\sum z = \sum_{v=1}^{N} v - N\frac{N+1}{2} = 0$; and $\sum xy = (\sum_j x_j)(\sum_i y_i) = 0$. (2) $\mathrm{Var}(x) = \frac{1}{n}\sum_j x_j^2$ is the variance of n equally spaced points with unit spacing, $(n^2-1)/12$; the same holds for y, and for z with N points. (3) Since $M(n) = n\frac{N+1}{2}$, $\sum_{c\in L} z_c = \sum_{c\in L} s_c - M(n) = \delta_L$. The cells of a row i have $y = y_i$ and x running over all x_j (sum 0); for a column the roles are exchanged; each diagonal is invariant under $(x,y)\mapsto(-x,-y)$, so its position sums vanish. □

Figure 2 illustrates Lemma 2.2(3) for Dürer's square: along each of four lines the heights stack head to tail and return to zero, so the vector sum of the line is horizontal, $(0,6,0) = 4\,(0,\frac{3}{2},0)$ for the top row; after swapping two entries the stack of the affected lines fails to close, by the deviation δ_L.

Example 2.3 (Lo Shu)**.** For the Lo Shu square $S = \begin{pmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{pmatrix}$, $n = 3$ and the centre value is 5, so the nine points are

$$\begin{array}{lll} \mathbf{p}_{00} = (-1,1,-3), & \mathbf{p}_{01} = (0,1,2), & \mathbf{p}_{02} = (1,1,1), \\ \mathbf{p}_{10} = (-1,0,4), & \mathbf{p}_{11} = (0,0,0), & \mathbf{p}_{12} = (1,0,-4), \\ \mathbf{p}_{20} = (-1,-1,-1), & \mathbf{p}_{21} = (0,-1,-2), & \mathbf{p}_{22} = (1,-1,3). \end{array}$$

The top row has heights $-3+2+1=0$ and horizontal vector sum $(0,3,0)$. The centre cell maps to the origin, which is the only non-vertex point and is strictly interior (Proposition 3.1); the gem has f-vector $(8,16,10)$ and volume 18 (Figure 3).

Proposition 2.4 (D_4-invariance)**.** *Let $g \in D_4$ act on the grid and let gS be the arrangement with $(gS)_c = s_{g^{-1}c}$. Then $\mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S)$, where $\hat{g}(x,y,z) = (g(x,y),z)$ is the isometry of $\mathbb{R}^3$ that applies the orthogonal map g of the (x,y)-plane and fixes z;* $\det\hat{g} = \det g$*. In particular $\mathcal{G}(gS)$ and $\mathcal{G}(S)$ are congruent, and the gem is an invariant of the D_4-orbit of S.*

Proof. In the centred coordinates (1), each $g \in D_4$ acts on the cell positions as an orthogonal map of the plane (a rotation by a multiple of 90° or a reflection in a coordinate axis or a diagonal), and $(x_{gc}, y_{gc}) = g(x_c, y_c)$. The point of gS over the cell gc has height $(gS)_{gc} - \frac{N+1}{2} = z_c$. Hence the point set of gS is $\{(g(x_c,y_c),z_c)\}_c = \hat{g}\{\mathbf{p}_c\}_c$, and an affine bijection maps the convex hull of a set onto the convex hull of its image. □

Magic squares are mapped to magic squares by D_4 (lines go to lines), so every *essentially different* magic square, that is, every D_4-orbit, has a single gem up to congruence.

3 Balance: what the vectors from the centroid do and do not say

Unweighted balance holds for every arrangement.

Proposition 3.1 (Centroid)**.** *For every arrangement S of order n, $\sum_{i,j} \mathbf{p}_{ij} = \mathbf{0}$: the origin is the centroid of the N points, each of which is a vector from it. Moreover $\mathbf{0}$ lies in the interior of $\mathcal{G}(S)$.*

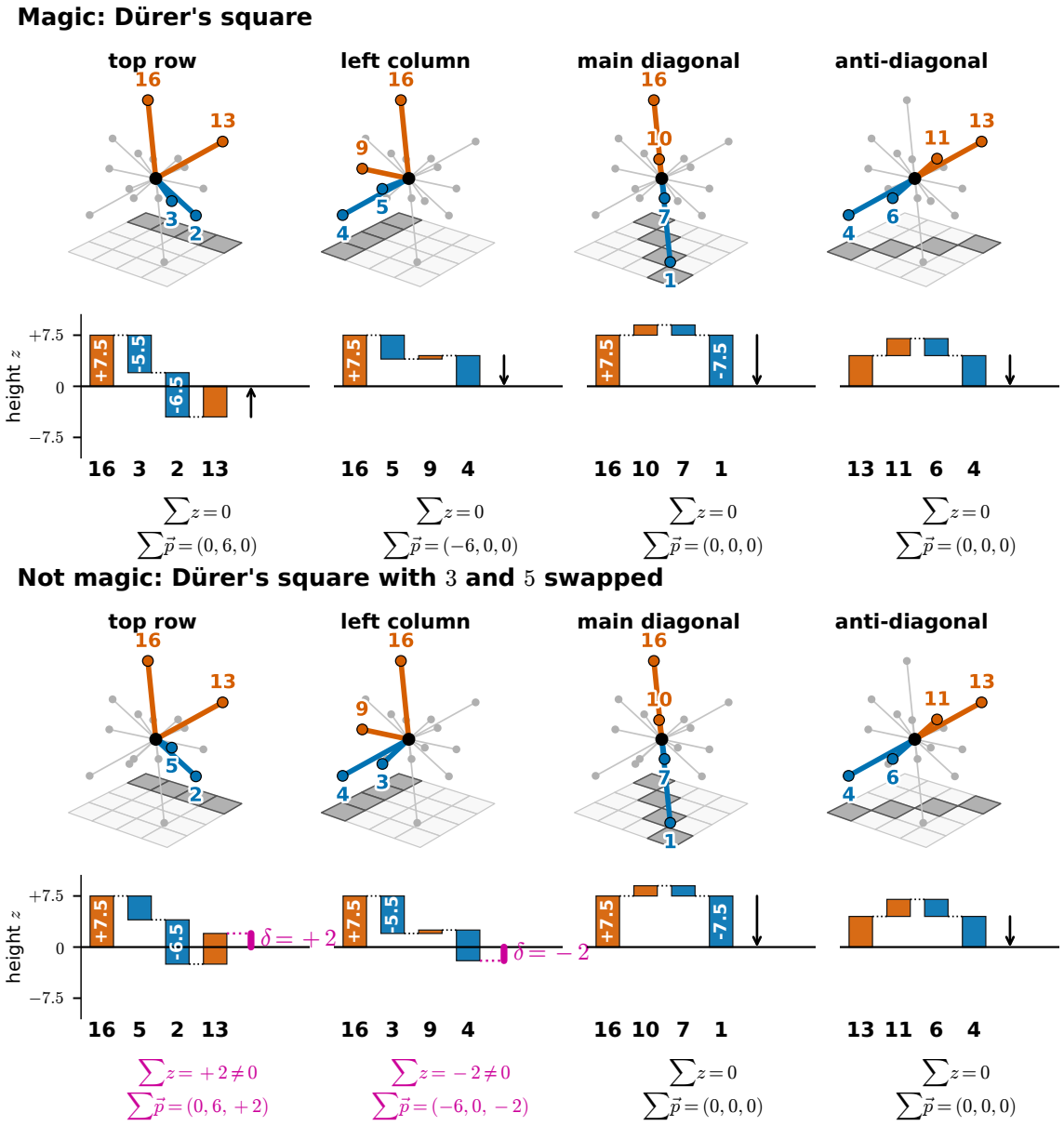


Figure 2: Magic means that heights cancel along every line. *Top:* for Dürer's square, the four vectors of the top row, left column, main diagonal and anti-diagonal (coloured, with their entries; the other twelve in grey; the cells of the line are shaded on the floor) and below each the heights $z = s - 8.5$ of the line stacked head to tail in line order; the running total returns to 0, so $\sum_{c \in L} \mathbf{p}_c$ is the horizontal vector $(n\bar{x}_L, n\bar{y}_L, 0)$ of Lemma 2.2(3), here $(0, 6, 0)$, $(-6, 0, 0)$, $\mathbf{0}$, $\mathbf{0}$. *Bottom:* the same after swapping the entries 3 and 5, which is not magic: the top row and the left column no longer close (deviations $\delta_L = +2$ and -2, in magenta) and their vector sums acquire the vertical component δ_L; the diagonals do not contain the swapped cells and still close.

Proof. The sum vanishes by Lemma 2.2(1). The centroid of a finite point set that is not contained in a plane is an interior point of its convex hull, as a convex combination with all weights positive; the points $\mathbf{p}_{ij}$ span $\mathbb{R}^3$ since the positions (x_{ij}, y_{ij}) do and the heights are not all equal for $n \geq 2$. □

This is a consequence of centring the coordinates and carries no information about S; in particular it does not distinguish magic squares from arbitrary arrangements. It is a statement about the point set with unit weights, and not about the solid: the hull depends only on its vertex set, points that are not vertices do not affect it, and the centre of mass of the solid $\mathcal{G}(S)$ need not be the origin (for 829 of the 880 gems of order four it is not). The hull records the extent of the point set (extreme points, faces, volume), not how the heights are distributed over the cells.

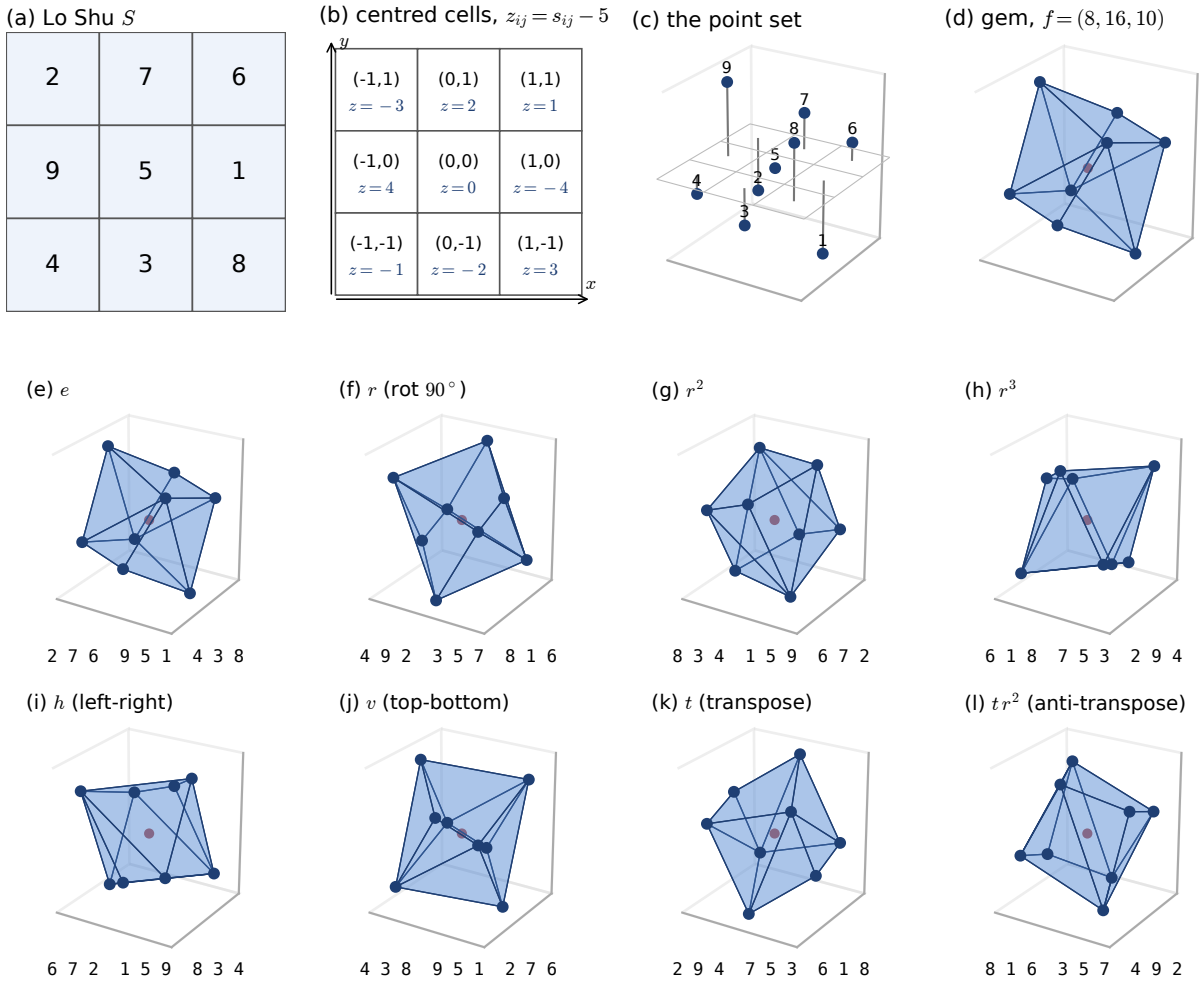


Figure 3: Construction of the Magic Gem and its D_4-invariance. Top row: (a) the Lo Shu square S; (b) the centred cells (x_{ij}, y_{ij}) with the heights $z_{ij} = s_{ij} - 5$; (c) the nine points $\mathbf{p}_{ij}$ over the grid, labelled by the entries s_{ij}; (d) their convex hull, the gem of Example 2.3, with f-vector $(8, 16, 10)$; the centre point $(0, 0, 0)$ is the only non-vertex point and is strictly interior. Rows two and three (e–l): the eight images of the Lo Shu under D_4—the identity e, the rotations r, r^2, r^3 by $90°, 180°, 270°$, the left–right and top–bottom reflections h, v, and the two diagonal reflections (transpose t and anti-transpose tr^2)—each printed row by row beneath its gem. All eight gems are drawn from the same viewpoint; by Proposition 2.4 they are the images of one another under the isometries $\hat{g}$, so they are congruent, and all have f-vector $(8, 16, 10)$ and volume 18.

Height-weighted balance is what magic means. Take the *signed mass* $m_{ij} = z_{ij}$ at the planar position (x_{ij}, y_{ij}), with total mass 0. Write $\rho_i = \sum_j z_{ij}$ and $\gamma_j = \sum_i z_{ij}$ for the row and column mass sums; by Lemma 2.2(3) they are the row and column deviations of S.

Proposition 3.2 (Height-weighted moments)**.** *Let S be an arrangement of order $n \geq 3$.*

1. *S is semi-magic if and only if $\rho_i = 0$ for every row and $\gamma_j = 0$ for every column.*
2. *For every function $\varphi\colon \mathbb{R} \to \mathbb{R}$,*

$$\sum_{i,j} z_{ij}\, \varphi(x_{ij}) = \sum_j \varphi(x_j)\gamma_j, \qquad \sum_{i,j} z_{ij}\, \varphi(y_{ij}) = \sum_i \varphi(y_i)\rho_i.$$

Hence if S is semi-magic, $\sum z\, \varphi(x) = \sum z\, \varphi(y) = 0$ for every φ (for $\varphi(t) = t^k$, all $k \geq 0$; for $k = 1$ the height-weighted centroid is the origin). Conversely, if $\sum z\, \varphi(x) = \sum z\, \varphi(y) = 0$ for all φ, then S is semi-magic.

3. The mixed moment is not forced: for $n = 4$ there are magic squares with $\sum z\,xy \neq 0$.

Proof. (1) By Lemma 2.2(3), a row sums to $M(n)$ if and only if its heights sum to 0; likewise for columns. (2) Grouping the cells by column, $\sum_{i,j} z_{ij}\varphi(x_j) = \sum_j \varphi(x_j)\gamma_j$; the row identity is the same with y_i, ρ_i. If $\gamma = \rho = 0$ both vanish. Conversely take φ to be the indicator of a single value x_{j_0} (the x_j are distinct): then $\gamma_{j_0} = 0$; likewise $\rho_{i_0} = 0$, and (1) applies. (3) Since $\sum xy = 0$, $\sum z\,xy = \sum s\,xy = -P_{xy}$, the product of inertia of Loly's value-weighted planar body. Exact enumeration over the 880 essentially different magic squares of order 4 gives $|P_{xy}| \in \{0, \ldots, 60\}$ with 48 distinct values; it vanishes for 56 classes and is non-zero for the other 824. □

Figure 4 shows both halves of Proposition 3.2. In panel a the entries themselves are piles of checkers on a checkerboard, s_{ij} checkers on cell (i, j): every row and column carries 34 checkers, so the horizontal centre of mass of all the checkers lies over the centre O ($\sum s\,x = \sum s\,y = 0$, which for $n = 4$ is the height-weighted statement because $\sum x = \sum y = 0$). Hence each column of heights has zero total mass and the lever of signed masses balances, while the mixed moment $\sum z\,xy$ takes the value 0 for Dürer's square and +60 for another magic square. The same weighted balance in vector form is Figure 5.

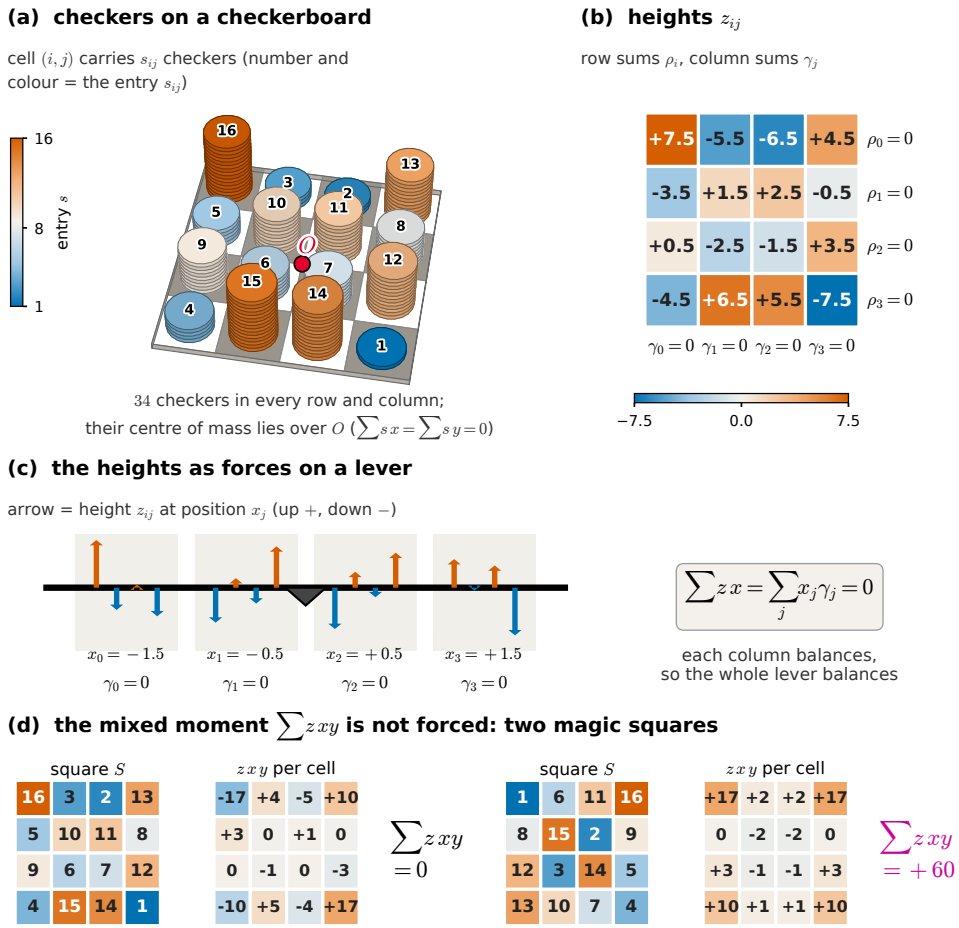


Figure 4: Height-weighted moments, order $n = 4$. (a) Dürer's square as checkers on a 4×4 checkerboard, seen from the front: the pile on cell (i, j) has s_{ij} checkers (labelled, and coloured by the entry s_{ij} with the colour bar shown); every row and column carries 34 checkers, and the centre of mass of all the checkers lies over the board centre O (red), $\sum s\,x = \sum s\,y = 0$. (b) The heights with row sums ρ_i and column sums γ_j, all zero. (c) Reading z_{ij} as a force at position x_j on a lever (up +, down −): every column has net force $\gamma_j = 0$, hence $\sum z\,x = \sum_j x_j\gamma_j = 0$ (Proposition 3.2(2)); the same holds for y and the rows. (d) The mixed moment is not forced: for two magic squares, the heights z and the cell contributions $z\,x\,y$ (rounded to integers), which sum to 0 for Dürer's square and to +60 for the square $\begin{smallmatrix} 1 & 6 & 11 & 16 \\ 8 & 15 & 2 & 9 \\ 12 & 3 & 14 & 5 \\ 13 & 10 & 7 & 4 \end{smallmatrix}$ (so $P_{xy} = -60$).

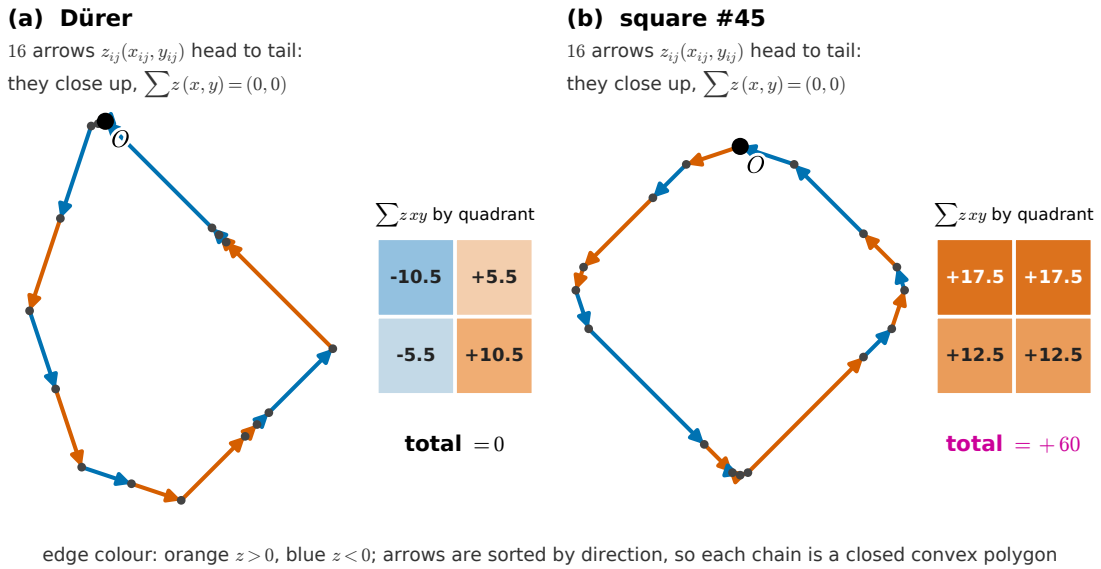


Figure 5: The height-weighted vector sum $\sum_{i,j} z_{ij}\,(x_{ij}, y_{ij}) = (0,0)$ for two magic squares: Dürer's square and square #45 of the 880 (the second square of Figure 4d). The 16 moment arms $z_{ij}(x_{ij}, y_{ij})$ (orange for $z > 0$, blue for $z < 0$) are laid head to tail in order of direction; the chain closes at O in both cases, which is $\sum z\,x = \sum z\,y = 0$ (Proposition 3.2(2)). The right-hand tables give $\sum z\,xy$ for the four quadrants of the grid; they add to 0 for Dürer's square and to +60 for square #45, so the weighted balance does not determine the mixed moment (Proposition 3.2(3)).

Thus the height-weighted statements guaranteed by the line sums are exactly the pure axial ones: the vanishing of the first moment along each axis and Loly's invariant below. The mixed moment can vanish for structural reasons: if S is associative, z is odd under $(x, y) \mapsto (-x, -y)$ (Section 4) and xy is even, so $\sum z\,xy = 0$ (this accounts for the order-3 square and for 48 of the 56 vanishing classes of order 4).

Corollary 3.3 (Loly's moment of inertia)**.** *For every arrangement,* $I(S) := \sum_{i,j} s_{ij}(x_{ij}^2 + y_{ij}^2) = \sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i + \frac{n^2(n^4-1)}{12}$. *In particular* $I(S) = n^2(n^4-1)/12$ *for every semi-magic square.*

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$. The first part is $\sum z\,(x^2 + y^2)$, equal to $\sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i$ by Proposition 3.2(2); the second is $\frac{N+1}{2}\sum(x^2+y^2) = \frac{N+1}{2}\cdot 2N\frac{n^2-1}{12} = \frac{n^2(n^4-1)}{12}$ by Lemma 2.2(2). □

This is Loly's invariance [12], which he derived from the row and column sums. For $n = 3, \ldots, 9$ the value is $60, 340, 1300, 3885, 9800, 21840, 44280$.

Remark 3.4 (Inertia tensor of the unit-mass point set). With unit masses at the N points, the centre of mass is the origin and the inertia tensor about it is $I_{jk} = N(\delta_{jk}\,\mathrm{tr}\,\Sigma - \Sigma_{jk})$, where Σ is the covariance matrix of the points [11]. By Lemma 2.2 and Proposition 3.2(2), $I_{xx} = I_{yy} = n^2(n^2-1)(n^2+2)/12$, $I_{zz} = n^2(n^2-1)/6$, $I_{xy} = 0$, $I_{xz} = -\sum_j x_j\gamma_j$ and $I_{yz} = -\sum_i y_i\rho_i$. The diagonal entries are the same for all $N!$ arrangements and the tensor is diagonal for every semi-magic square, so the inertia of the unit-mass point set does not depend on the square. The informative planar object is Loly's value-weighted body: its diagonal entries are fixed by the row and column sums (Corollary 3.3), whereas the product of inertia $P_{xy} = -\sum s_{ij}x_{ij}y_{ij}$ is a mixed moment, fixed by no line sum; D_4 changes at most its sign.

Remark 3.5 (Covariance determinant). Let Σ be the covariance matrix of (x, y, z) over the N points. Since $\Sigma_{xy} = 0$ and $\Sigma_{xx} = \Sigma_{yy} =: a$, $\Sigma_{zz} =: c$ are the same for all arrangements (Lemma 2.2),

$$\det\Sigma = a\,a\,c - a\,\Sigma_{xz}^2 - a\,\Sigma_{yz}^2 = a\big(ac - \Sigma_{xz}^2 - \Sigma_{yz}^2\big),$$

with $N\Sigma_{xz} = \sum z\,x$ and $N\Sigma_{yz} = \sum z\,y$. So among all arrangements of order n the determinant, which is proportional to the squared volume of the covariance ellipsoid, is largest exactly when

$\sum z\,x = \sum z\,y = 0$, in particular for every semi-magic square (Proposition 3.2(2) with $\varphi(t) = t$). This concerns the point set, not its hull. Numerically ($n = 4$, 10^5 uniformly random arrangements, `anc/vertex_statistics.py`), the hull volume of a random arrangement has mean 75.7 and is positively rank-correlated with $\det\Sigma$ (Spearman 0.65), and the median of the 880 magic gem volumes (85.2; mean 84.05) lies at the 91st percentile of the random volumes, numerical statements about samples only.

4 Symmetry of the gems

Let $\rho(x, y, z) = (x, y, -z)$ be the reflection in the plane $z = 0$. The *complement* of an arrangement is $\overline{S} = N + 1 - S$ (so $\overline{S}$ has height function $-z$), and S is *associative* if $S + r^2 S = N + 1$, where r^2 is the half-turn of the grid; S is *self-complementary* if $\overline{S} = gS$ for some $g \in D_4$. Complements of magic squares are magic.

Lemma 4.1 (Complement is a mirror image)**.** *$\mathcal{G}(\overline{S}) = \rho\,\mathcal{G}(S)$. If $\overline{S} = gS$ with $g \in D_4$, then $\iota_g := \hat{g}^{-1}\rho$, i.e. $\iota_g(x, y, z) = (g^{-1}(x, y), -z)$, is an isometry of $\mathbb{R}^3$ mapping $\mathcal{G}(S)$ onto itself, with $\det \iota_g = -\det g$.*

Proof. The points of $\overline{S}$ are $(x_c, y_c, -z_c) = \rho\,\mathbf{p}_c$. If $\overline{S} = gS$ then $\rho\,\mathcal{G}(S) = \mathcal{G}(\overline{S}) = \mathcal{G}(gS) = \hat{g}\,\mathcal{G}(S)$ by Proposition 2.4, so $\hat{g}^{-1}\rho\,\mathcal{G}(S) = \mathcal{G}(S)$. □

Proposition 4.2 (Which elements of D_4 can complement)**.** *Let $\overline{S} = gS$ for an arrangement S of order $n \geq 2$. Then $g^2 = \mathrm{id}$ and g has no fixed cell other than possibly the centre cell of an odd grid, whose entry is then $(N + 1)/2$. Consequently:*

1. *for n odd, $g = r^2$, so S is associative;*
2. *for n even, $g \in \{r^2, h, v\}$ with h, v the left–right and top–bottom reflections; the diagonal reflections and the quarter-turns never occur.*

The corresponding isometries of $\mathcal{G}(S)$ are $\iota_{r^2}(\mathbf{p}) = -\mathbf{p}$ (point reflection, improper), $\iota_h(x, y, z) = (-x, y, -z)$ (half-turn about the y-axis, proper) and $\iota_v(x, y, z) = (x, -y, -z)$ (half-turn about the x-axis, proper).

Proof. Applying g to $\overline{S} = gS$ gives $g\overline{S} = \overline{gS} = S$, so $g^2 S = \overline{\overline{S}} = S$; since the entries are distinct, g^2 fixes every cell, hence $g^2 = \mathrm{id}$ (D_4 acts faithfully on the cells for $n \geq 2$). If g fixes a cell c then $s_c = N + 1 - s_c$. The involutions of D_4 are r^2, h, v and the two diagonal reflections. The diagonal reflections fix $n \geq 2$ cells and are excluded; h and v fix the n cells of the middle column (row) when n is odd, and none when n is even; r^2 fixes only the centre cell when n is odd. For n odd the centre cell is the only cell that can have the value $(N + 1)/2$, so $g = r^2$. The formulas for ι_g follow from Lemma 4.1, and the determinants are $-\det r^2 = -1$, $-\det h = -\det v = +1$. □

By Proposition 4.2, associative squares give gems with a point reflection: *centrally symmetric* gems. (Every Siamese square is associative, which explains the symmetric vertex list in Proposition 5.3; Dürer's square is associative and its gem is centrally symmetric, Figure 6a, Figure 7a.) Self-complementary squares that are not associative give gems with a half-turn symmetry. For order four we determine the complete isometry group of every gem. For a finite set $X \subset \mathbb{R}^3$ let $\mathrm{Iso}(X)$ be the group of isometries of $\mathbb{R}^3$ mapping X to itself.

Theorem 4.3 (Isometry groups of the order-four gems; computer-assisted)**.** *Let S be one of the 880 essentially different magic squares of order 4 and $\mathcal{G} = \mathcal{G}(S)$. Then $\mathrm{Iso}(\mathcal{G}) = \mathrm{Iso}(V) = \mathrm{Iso}(\{\mathbf{p}_{ij}\})$, where V is the vertex set, and:*

1. *S is associative (48 squares, Dudeney's group III) if and only if $\mathrm{Iso}(\mathcal{G}) = \{\mathrm{id}, -\mathrm{id}\}$ (the group C_i);*

2. *S is self-complementary but not associative (*304 *squares, Dudeney's group VI) if and only if* $\mathrm{Iso}(\mathcal{G}) = \{\mathrm{id}, \iota\}$ *with* ι *a half-turn about a horizontal coordinate axis (the group* C_2*);*

3. *for the remaining* 528 *squares* $\mathrm{Iso}(\mathcal{G})$ *is trivial.*

Consequently $\mathcal{G}(S)$ *has an improper isometry if and only if* S *is associative; the* 832 *gems of the cases (2) and (3) are chiral; and the congruence classes of the* 880 *gems number* $48{+}304{+}264 = 616$*, the* 528 *gems of case (3) forming* 264 *pairs* $\{\mathcal{G}(S), \mathcal{G}(\overline{S})\}$ *of mirror images in the plane* $z = 0$*.*

Proof. The inclusions "$\supseteq$" and the identification of the generators for the associative and self-complementary squares are Lemma 4.1 and Proposition 4.2; the classes (1), (2) are exactly the self-complementary ones, which were counted as 48 and 304 (all of the form h or v after Proposition 4.2) by exact comparison of the 8 images of S with $\overline{S}$. For the reverse inclusions we computed, for each of the 880 squares, *all* isometries of its vertex set and of its 16-point set, in exact integer arithmetic: the vertex set is centred at its centroid (scaled to integers), the images of three linearly independent vertices are enumerated among the vertices with the same Gram matrix entries, each candidate orthogonal map is accepted only if the induced bijection of the vertices preserves all pairwise integer squared distances (Section A). This yields the group orders $2, 2, 1$ in the three cases, equal for the vertex set and the 16-point set; since $\mathcal{G}$ is the convex hull of V, $\mathrm{Iso}(\mathcal{G}) = \mathrm{Iso}(V)$ (an isometry preserves extreme points). A gem is chiral if it has no improper isometry; the group C_2 consists of proper maps, which gives the chirality statement. Finally, congruent gems come from the same complement orbit (Proposition 5.1), and in case (3) the classes of S and $\overline{S}$ are distinct, giving 264 pairs. □

Table 1: Symmetry types of the 880 order-four gems. V: number of hull vertices; "classes" counts congruence classes of gems.

type of S	Dudeney	squares	$\mathrm{Iso}(\mathcal{G})$ (order)	classes	V
associative	III	48	C_i, inversion (2)	48	$10, 12, 14, 16$
self-complementary, not associative	VI	304	C_2, half-turn (2)	304	$8, 10, 12, 14, 16$
neither	others	528	trivial (1)	264	$8, \ldots, 16$
total		880		616	

Proposition 4.4 (Facet orbits)**.** *Let* S *be one of the* 880 *squares and let* $\mathrm{Iso}(\mathcal{G})$ *act on the facets of* $\mathcal{G}$*.*

1. *If* S *is associative, every facet* F *has a distinct antipodal partner* $-F$*, a facet in the parallel plane on the other side of the origin; so there are no fixed facets and the facets form* $F/2$ *antipodal pairs. Here* $F \in \{12, 14, 16, 18, 20, 22\}$ *with* $1, 12, 12, 17, 4, 2$ *gems respectively, and* $V \in \{10, 12, 14, 16\}$ *with* $13, 28, 4, 3$ *gems. The partner* $-F$ *has the opposite outward normal; as plane polygons* F *and* $-F$ *are congruent by a rotation through* π*, although the point reflection carrying* F *to* $-F$ *is improper.*

2. *If* S *is self-complementary and not associative, with half-turn* ι *about the* y*-axis (say), the facet orbits have size* 1 *or* 2*. A facet is fixed if and only if it lies in* $y = \pm\frac{3}{2}$*; such a facet is a parallelogram (the hull of the top or bottom row), and exists unless the four points of that row are collinear. There are two fixed facets for* 282 *gems, one for* 18 *and none for* 4*.*

3. *The vertices of a gem with a symmetry fall into orbits of size* 2*, so* V *is even for the* 352 *self-complementary squares.*

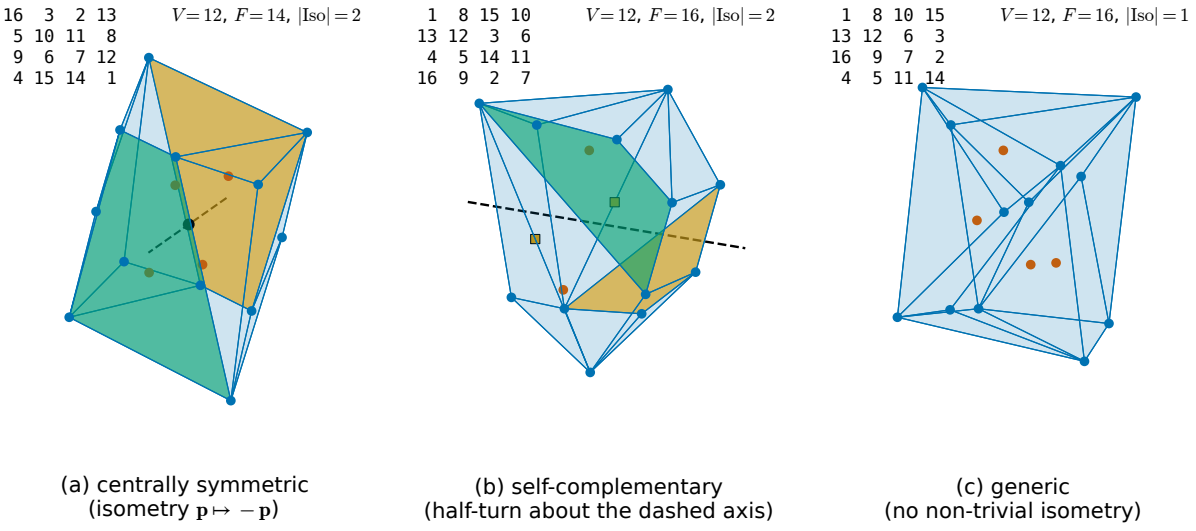
Proof. Both nontrivial isometries fix the origin, which is an interior point of $\mathcal{G}$ (Proposition 3.1), so no facet plane $\{\mathbf{n} \cdot \mathbf{p} = d\}$ has $d = 0$. (1) $-F \neq F$ because $-$ maps the facet plane $\mathbf{n} \cdot \mathbf{p} = d$ to $\mathbf{n} \cdot \mathbf{p} = -d \neq d$. (2) The half-turn $\iota = (x, y, z) \mapsto (-x, y, -z)$ maps the plane $\mathbf{n} \cdot \mathbf{p} = d$ to

$\iota\mathbf{n}\cdot\mathbf{p} = d$, which is the same plane iff $\iota\mathbf{n} = \mathbf{n}$, i.e. $\mathbf{n} \parallel (0,1,0)$; the facet planes with this normal are $y = \pm\frac{3}{2}$. The hull of the top row (the points with $y = \frac{3}{2}$) is ι-invariant; its vertices come in pairs $\pm(x,z)$ and a convex polygon with at most four vertices and a centre of symmetry is a segment or a parallelogram. The counts are those of the exact computation. (3) ι and $-\mathrm{id}$ fix no vertex: the half-turn fixes only points with $x = z = 0$, while x is a half-odd integer; the point reflection fixes only the origin. □

Proposition 4.5 (Parity of the point types for associative squares)**.** *Let S be associative of order n and let V, I, B be the numbers of vertices, strictly interior points and boundary non-vertex points of $\mathcal{G}(S)$. Then V and B are even, and $I \equiv n \pmod 2$. In particular for even n all three numbers are even.*

Proof. Associativity says that the cell c' opposite to c carries $s_{c'} = N + 1 - s_c$, so $\mathbf{p}_{c'} = -\mathbf{p}_c$: the point set is invariant under the linear isometry $-\mathrm{id}$, which therefore maps $\mathcal{G}(S)$ to itself, and permutes its vertices, its strictly interior points and its boundary non-vertex points among themselves. The only fixed point of $-\mathrm{id}$ is the origin, which is a point $\mathbf{p}_{ij}$ only if n is odd (the central cell has $s = \frac{N+1}{2}$), and then it is strictly interior by Proposition 3.1. Hence the involution is fixed-point free on vertices and on boundary non-vertex points, and on strictly interior points when n is even; for odd n it fixes exactly one of the latter. □

For $n = 4$ the 48 associative gems have $V \in \{10, 12, 14, 16\}$ and $I \in \{0, 2, 4\}$ (4, 23 and 21 gems), in accordance with Proposition 4.5 (exact computation, `anc/box_and_classes.py`); among the non-associative squares both parities of I occur.



(a) centrally symmetric (isometry $\mathbf{p} \mapsto -\mathbf{p}$)

(b) self-complementary (half-turn about the dashed axis)

(c) generic (no non-trivial isometry)

Figure 6: Symmetry types of order-four gems (same viewpoint; strictly interior points are red discs, boundary non-vertex points orange squares; the square is printed above each gem). (a) Dürer's square: a centrally symmetric gem; an antipodal pair of facets F (orange) and $-F$ (green) and the segment joining their centroids through the centre $\mathbf{0}$ (dashed). (b) A self-complementary, non-associative square (square #124): the half-turn ι about the dashed axis swaps the highlighted pair of facets and has no other symmetry. (c) A square with no symmetry (square #102): the gem is chiral, and its mirror image in $z = 0$ is the gem of the complement $17 - S$.

Figure 7 shows the same three squares as point sets in the height picture: for the associative square the 16 points form 8 antipodal pairs $\{\mathbf{p}, -\mathbf{p}\}$ whose segments all pass through O; for the self-complementary square the half-turn about the dashed horizontal axis exchanges the 8 pairs, each segment meeting the axis perpendicularly at its midpoint; the third square has no such pairing, in agreement with Theorem 4.3.

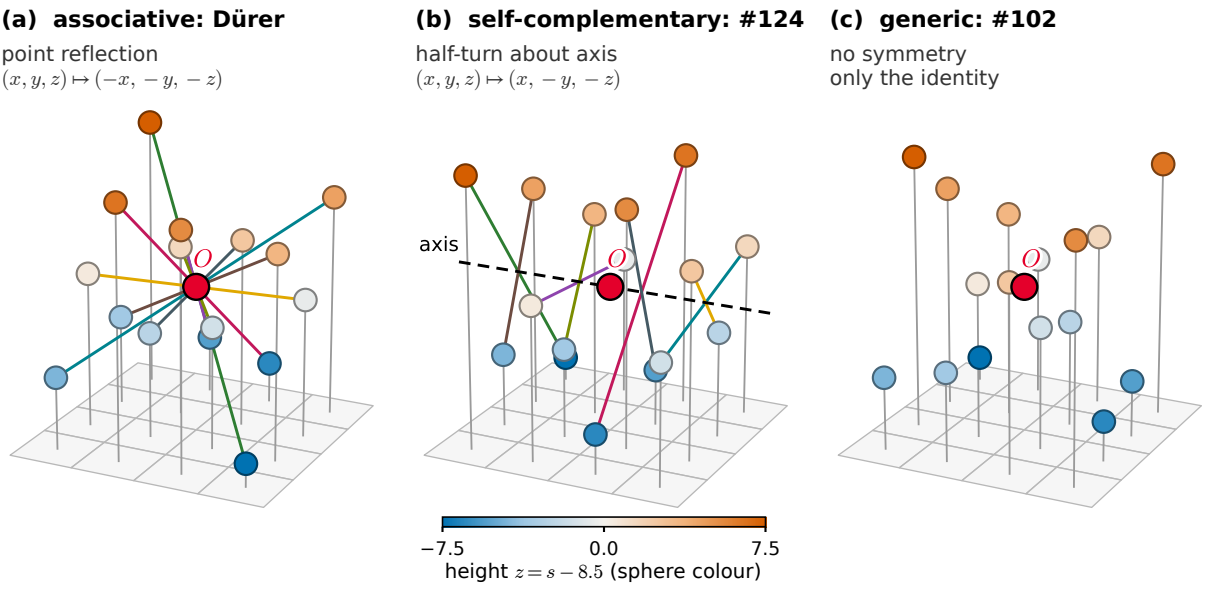


Figure 7: The 16 points $\mathbf{p}_{ij}$ as spheres on stems over the floor grid, coloured by height z (same squares and viewpoint as Figure 6; O red). (a) Dürer's square: the points form 8 antipodal pairs $\{\mathbf{p}, -\mathbf{p}\}$, joined by segments through O. (b) Square #124: the half-turn $(x, y, z) \mapsto (x, -y, -z)$ about the dashed x-axis exchanges 8 pairs of points, each pair joined by a segment perpendicular to the axis; this gem is chiral. (c) Square #102: no non-trivial isometry preserves the point set.

Remark 4.6 (Classes of order-four squares and vertex counts; exact computation). Dudeney's group I consists exactly of the 48 pandiagonal squares (all broken diagonals sum to 34), group III of the 48 associative squares, and group VI of the 304 self-complementary non-associative squares, whose gems have the half-turn symmetry of Theorem 4.3; no order-four square is both pandiagonal and associative. The gems of the 48 pandiagonal squares have 11 to 15 vertices ($10, 20, 12, 4, 2$ gems with $11, \dots, 15$), never 16 and never at most 10, and each has between one and four strictly interior points. The 7 gems with 8 vertices are squares #50, #54, #672 (group VI) and #276, #277, #494, #496 (group VIII); none is pandiagonal or associative, and each has $I = B = 4$.

Remark 4.7 (Central-symmetry defect). For a convex body $K \subset \mathbb{R}^3$ let $\sigma(K) = \operatorname{vol}\big(\frac{1}{2}(K - K)\big)/\operatorname{vol}(K)$. The Brunn–Minkowski inequality gives $\operatorname{vol}(K - K)^{1/3} \geq \operatorname{vol}(K)^{1/3} + \operatorname{vol}(-K)^{1/3} = 2\operatorname{vol}(K)^{1/3}$, that is $\sigma(K) \geq 1$, and for bodies with non-empty interior equality holds if and only if K and $-K$ are homothetic [21]; since they have equal volume the homothety is a translation, so $\sigma(K) = 1$ if and only if K is centrally symmetric. The defect σ is invariant under all isometries including reflections, so $\sigma(\mathcal{G}(S)) = \sigma(\mathcal{G}(\overline{S}))$ and σ is constant on D_4-orbits. For the 880 order-four gems, σ was computed in exact rational arithmetic from the convex hull of the 256 pairwise differences of the points (`anc/central_symmetry_defect.py`): $\sigma = 1$ for exactly the 48 associative gems, and $\sigma \in [1057/1018, 145/88] \approx [1.038, 1.648]$ for the other 832, the extremes occurring for squares #569 and #218.

5 Exact geometry: the order-four gems and the Siamese gems

All hull computations in this section are exact (integer or rational arithmetic); the methods are in Section A, the scripts and data in the ancillary directory `anc/`.

5.1 Geometry of the 880 order-4 gems

There are 8 magic squares of order 3, all images of the Lo Shu under D_4, and 7,040 of order 4, forming 880 D_4-orbits; the order-four squares are attributed to Frénicle de Bessy (his list was published in 1693) and have been recounted by computer [4]. Dudeney sorted them into twelve

groups according to the pattern of complementary pairs $(a, 17-a)$ [8]; we use his groups only as a classical reference partition. The number 275,305,224 of essentially different order-five squares is due to Schroeppel (1973) [10]; for $n \geq 6$ exact counts are, to the author's knowledge, not available (see Trump's compilation [24]).

For each of the 880 essentially different order-4 magic squares we computed the Magic Gem $\mathcal{G}(S)$ exactly (integer facet planes, merged coplanar triangles, rational volumes; Section A); Euler's relation holds for all 880 gems. Figure 8 and Table 2 summarise the results.

Table 2: Face structure of the 880 order-4 gems by number of hull vertices V: number of squares, of distinct combinatorial types (face lattices up to isomorphism), of distinct exact volumes, and of distinct f-vectors (V, E, F). The full list of the 39 f-vectors with multiplicities is in the ancillary file `anc/results_geometry.json`.

V	8	9	10	11	12	13	14	15	16	all
squares	7	10	103	80	380	106	162	14	18	880
types	4	1	27	32	129	52	85	7	15	352
volumes	4	5	51	33	111	40	65	6	14	155
f-vectors	4	1	6	5	7	5	5	3	3	39

Vertex counts, f-vectors, volumes. Between 8 and all 16 of the points are hull vertices; $V = 12$ is modal (380/880). There are 39 distinct f-vectors, from $(8, 14, 8)$ (one gem, square #54) and $(8, 16, 10)$ (one gem) to $(16, 36, 22)$ (nine gems), the modal one being $(12, 26, 16)$ (164 gems); the multiset of facet sizes (number of triangles, quadrilaterals, ...) is determined by the f-vector in every case. The exact volume takes 155 distinct values, from 58 (square #427) to 292/3 (square #379), mean 84.05. The denominators are $1, 2, 3, 6$: all three coordinates of every point are half-odd integers when n is even, so the translate of a gem by $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope of $\mathbb{Z}^3$ and its volume lies in $\frac{1}{6}\mathbb{Z}$. Surface areas range from 116.5 to 148.6.

Combinatorial types. Two gems have the same combinatorial type if their vertex–facet incidence structures (equivalently, their face lattices) are isomorphic. Grouping by a Weisfeiler–Lehman hash of the bipartite vertex–facet graph and confirming every group by an explicit graph isomorphism test gives

352 combinatorial types

among the 880 gems (the hash was already exact: no hash class had to be split). Their sizes are very uneven; the number of types with $1, 2, 3, \ldots$ members is

squares per type	1	2	3	4	5	6	7	8	10	13	14	16	26	32
types	93	201	8	20	4	7	4	5	5	1	1	1	1	1

(Figure 8c); the largest type, with $f = (10, 22, 14)$ (12 triangles, 2 quadrilaterals), is shared by 32 squares, and the second largest, $f = (12, 24, 14)$, by 26. The 7 gems with 8 vertices fall into 4 types, including a single $(8, 14, 8)$ gem with four triangular and four quadrilateral faces (square #54, Fig. 8d) and four $(8, 18, 12)$ gems sharing one simplicial type with 12 triangular faces. The 18 gems with all 16 points extreme belong to 15 types.

Congruence classes and injectivity. Two gems are *congruent* if an isometry of $\mathbb{R}^3$ (rotations, reflections, translations) maps one onto the other; since a distance-preserving bijection between finite point sets extends to an isometry, congruence is decided exactly by testing for a bijection of hull-vertex sets preserving all integer squared distances (Section A).

Proposition 5.1 (class → gem)**.** *1. The* 880 *hull-vertex sets are pairwise distinct: the map from D_4-classes of order-*4 *magic squares to hull-vertex sets (as subsets of $\mathbb{R}^3$) is injective. (For the full* 16*-point sets this is trivial, since the point set determines the arrangement.)*

2. *Up to congruence there are exactly* 616 *gems. The congruence classes are precisely the orbits of the complement map* $S \mapsto 17-S$*, which acts on gems as the reflection* $z \mapsto -z$*:* 352 *squares are self-complementary (their gems have a non-trivial isometry, Theorem 4.3) and the other* 528 *form* 264 *mirror pairs. No two gems of non-complementary squares are congruent, not even as full* 16*-point sets.*

Part (2) is the only failure of injectivity, and it is the expected one: $17 - S$ is magic whenever S is, and its gem is the mirror image of $\mathcal{G}(S)$ in the plane $z = 0$. The 352 self-complementary classes are exactly Dudeney's group III (the 48 associative squares) together with all 304 squares of group VI [8]. The coincidence $352 = \#\{\text{types}\} = \#\{\text{self-complementary squares}\}$ is numerical only: the two partitions of the 880 are different.

Dudeney groups. Crossing the hull-vertex count with Dudeney's twelve groups (table in the ancillary files) shows that geometry is only loosely tied to Dudeney type: every group with more than eight members spans at least three vertex counts and at least 20 combinatorial types; the 8-vertex gems occur only in groups VI (three) and VIII (four).

The groups III and VI, the self-complementary ones, have only even vertex counts; this is Proposition 4.4(3).

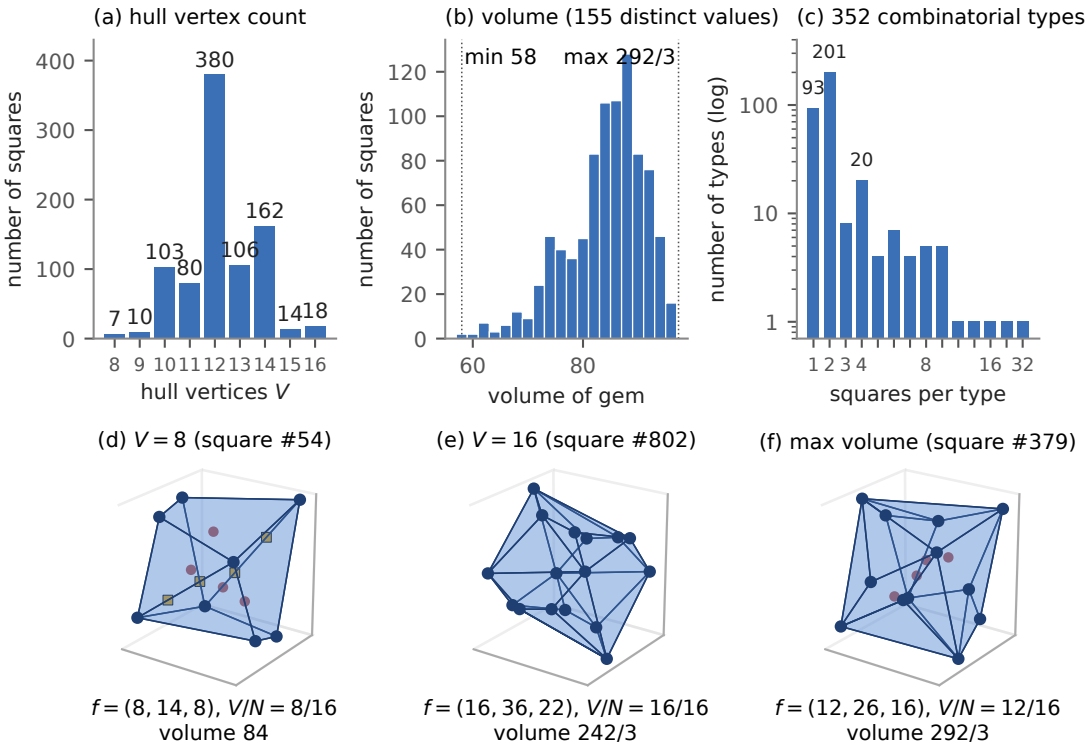


Figure 8: Geometry of the 880 order-4 Magic Gems. (a) Number of hull vertices (8 to 16; modal value 12). (b) Exact hull volume (155 distinct values between 58 and 292/3). (c) Sizes of the 352 combinatorial types (logarithmic scale): 93 types occur once, 201 twice, and one type is shared by 32 squares. (d)–(f) Three extreme gems, drawn with translucent faces after merging coplanar triangles: the unique $(8, 14, 8)$ gem (square #54; its four strictly interior points are red discs and its four boundary non-vertex points orange squares), a gem with all 16 points extreme (square #802, Dürer's square with its first two rows exchanged and its last two rows exchanged), and the gem of largest volume (square #379).

Named squares. The gems of the Lo Shu, of Dürer's square and of the Siamese square of order 5 have hull fractions 8/9, 12/16 and 12/25 and exact volumes 18, 92 and 830/3. Dürer's own gem is centrally symmetric with twelve vertices, but exchanging its first two rows and its last two rows (which preserves the magic property) gives a square whose gem has all sixteen points extreme

(square #802, Figure 8e): the hull-vertex count is a property of the square, not of the Dudeney group or of the construction method.

5.2 Extreme points and the bounding box

Let $B_n = [-\frac{n-1}{2}, \frac{n-1}{2}]^2 \times [-\frac{N-1}{2}, \frac{N-1}{2}]$, a box of volume $(n-1)^2(N-1)$ that depends only on n. The *side lines* of the grid are its first and last rows and columns.

Proposition 5.2 (Bounding box and forced vertices)**.** *Let S be any arrangement of order $n \geq 2$.*

1. *$\mathcal{G}(S) \subseteq B_n$.*
2. *The points carrying the entries 1 and N are the only points on the planes $z = \mp\frac{N-1}{2}$, and they are vertices of $\mathcal{G}(S)$.*
3. *The points over the four corner cells of the grid are vertices.*
4. *For each of the four side lines, the points over its cells carrying the smallest and the largest entry of that line are vertices.*

Proof. (1) $|x_{ij}|, |y_{ij}| \leq \frac{n-1}{2}$ and $1 \leq s_{ij} \leq N$, and B_n is convex. A point that is the unique maximiser over $\{\mathbf{p}_{ij}\}$ of a linear functional is a vertex. (2) The entries are distinct, so z is uniquely minimal where $s = 1$ and uniquely maximal where $s = N$. (3) On the grid the functional $-x + y$ attains its maximum $n - 1$ only at the top-left cell, and similarly for the other corners. (4) Take the left column, where $x = -\frac{n-1}{2}$ and the other columns have $x \geq -\frac{n-1}{2} + 1$. For $\varepsilon = 1/N$ the functional $-x + \varepsilon z$ has, on points of the column, values greater than $\frac{n-1}{2} - \frac{1}{2}$ and, on all other points, values at most $\frac{n-1}{2} - 1 + \frac{1}{2}$; so its maximiser lies in the column and is the point of largest height there, which is unique. Replacing εz by $-\varepsilon z$ gives the point of smallest height, and the other three sides are obtained by applying elements of D_4. □

The vertices named in (2)–(4) number at least 4 and at most 14; for $n = 4$ they number between 4 and 13 over the 880 gems, and nothing more is forced (exact computation, `anc/box_and_classes.py`): the only cells carrying a vertex in all 880 gems are the four corners, and the only entries that always label a vertex are 1 and 16. In particular 2 and $N - 1$ are not forced: in eight gems the point carrying 2 is a boundary non-vertex point, for instance for square #593, $\begin{smallmatrix} 4 & 1 & 16 & 13 \\ 5 & 15 & 2 & 12 \\ 14 & 8 & 9 & 3 \\ 11 & 10 & 7 & 6 \end{smallmatrix}$, and the complement $17 - S$ (the mirror image, Lemma 4.1) has the same property for 15. The twelve vertices of a Siamese gem (Proposition 5.3) are its forced vertices.

Table 3 gives the volume of the order-four gems as a fraction of $\operatorname{vol} B_4 = 135$ and the numbers I and B of strictly interior and boundary non-vertex points (exact computation). Gems fill between $58/135 \approx 43\%$ and $292/405 \approx 72\%$ of the box, on average $2219/3564 \approx 62\%$. At most four points are strictly interior, while one gem has $I = 0$, $B = 6$.

Table 3: Order-four gems: volume as a fraction of the box B_4 (exact; rounded), by number V of vertices, and numbers of gems with k strictly interior points ($I = k$) and with k boundary non-vertex points ($B = k$).

V	8	9	10	11	12	13	14	15	16	all
min	.622	.588	.481	.549	.430	.456	.452	.498	.449	.430
mean	.642	.623	.631	.630	.634	.614	.608	.560	.536	.623
max	.648	.667	.701	.699	.721	.688	.691	.615	.598	.721

k	0	1	2	3	4	6
gems with $I = k$	37	32	353	178	280	0
gems with $B = k$	357	204	257	40	21	1

5.3 Lattice structure and Ehrhart polynomials

For even n all coordinates of every $\mathbf{p}_{ij}$ are half-odd integers, so $Q(S) = \mathcal{G}(S) + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is a lattice polytope; any other lattice translate has the same Ehrhart polynomial $L_S(t) = \#(tQ \cap \mathbb{Z}^3) = \mathrm{vol}(\mathcal{G})\, t^3 + c_2 t^2 + c_1 t + 1$ [3], and L_S is the same for all images gS, $g \in D_4$, since $\hat{g}$ preserves $\mathbb{Z}^3$ up to an integer translation of Q. Writing $\sum_{k\geq 0} L_S(k)t^k = (h_0^* + h_1^* t + h_2^* t^2 + h_3^* t^3)/(1-t)^4$, the h_i^* are non-negative integers [23], $h_0^* = 1$, $\sum h_i^* = 6\,\mathrm{vol}$, and h_3^* is the number of interior lattice points of Q [3].

Exact computation (`anc/ehrhart_880.py`; lattice points of tQ counted for $t \leq 6$, the cubic interpolated from $t \leq 3$ and checked at $t = 4, 5, 6$, with h_3^* checked against the interior count and $\sum h_i^*$ against $6\,\mathrm{vol}$). Over the 880 gems, $h_1^* \in [70, 136]$, $h_2^* \in [233, 403]$ and $h_3^* \in [34, 52]$. There are 554 distinct Ehrhart polynomials, against 616 congruence classes, 352 combinatorial types, 39 f-vectors and 155 volumes: congruent gems (complement partners; the reflection $z \mapsto -z$ followed by an integer translation is a unimodular affine map of $\mathbb{Z}^3$) have equal polynomials, but non-congruent gems can share one, and L_S is determined neither by the f-vector (it takes several values on 29 of the 39 f-vectors), nor by the combinatorial type (81 of 352 types), nor by the volume (104 of 155 volumes). For the Lo Shu square (odd order: the gem itself is a lattice polytope) $L(t) = 1 + t + 13t^2 + 18t^3$ and $h^* = (1, 29, 73, 5)$; for Dürer's square $L(t) = 1 - 3t + 44t^2 + 92t^3$ and $h^* = (1, 130, 377, 44)$; the leading coefficients are the volumes 18 and 92.

5.4 Siamese gems have twelve vertices

For the Siamese (de la Loubère) squares of odd order the hull-vertex count does not grow with n. We computed the gems for $n = 3, 5, \ldots, 21$: the Lo Shu gem has $f = (8, 16, 10)$, and every Siamese gem with $n \geq 5$ has $f = (12, 26, 16)$ with 12 triangular and 4 quadrilateral facets, so the hull fraction $12/n^2$ tends to zero. This is a theorem, with a short computer-assisted proof.

Proposition 5.3 (Siamese gems)**.** *Let $n = 2m + 1 \geq 5$ and let S be the Siamese square of order n (1 in the middle cell of the top row, successive entries one step up and to the right modulo n, one step down when blocked). Then $\mathcal{G}(S)$ has exactly 12 vertices, 26 edges and 16 facets (12 triangles, 4 quadrilaterals). With 0-indexed cells (i, j) and $c = (n^2+1)/2$, the vertices are the points over the cells*

$(0,0)$: $c + m + 2$,	$(0, m-1)$: $n^2 - 1$,	$(0, m)$: 1,
$(0, 2m)$: $c + m$,	$(m-1, 0)$: $n^2 - m$,	$(m, 0)$: $m + 2$,
$(m, 2m)$: $n^2 - m - 1$,	$(m+1, 2m)$: $m + 1$,	$(2m, 0)$: $c - m$,
$(2m, m)$: n^2,	$(2m, m+1)$: 2,	$(2m, 2m)$: $c - m - 2$,

i.e. the four corners, and on each side of the square the two cells holding the side's extreme values. For $n = 3$ the same construction gives the Lo Shu gem, with 8 vertices (all cells except the centre).

Proof. Step 1: closed form. Write $\langle t \rangle \in \{0, \ldots, n-1\}$ for t mod n. We claim $s(i, j) = n\langle i + j - m \rangle + \langle i + 2j - 2m \rangle + 1$. Indeed, along a run of n consecutive entries the move $(i, j) \to (i-1, j+1)$ fixes $i + j$ and increases $i + 2j$ by 1 modulo n, while the "blocked" move $(i, j) \to (i+1, j)$ increases both $i + j$ and $i + 2j$ by 1; the entry 1 sits at $(0, m)$ where both expressions vanish. Hence by induction the entry k lies in a cell with $\langle i + j - m \rangle = \lfloor (k-1)/n \rfloor$ and $\langle i + 2j - 2m \rangle = \langle k - 1 \rangle$; since $(i, j) \mapsto (i + j,\, i + 2j)$ is a bijection of $\mathbb{Z}_n^2$ (determinant 1), this determines the cell, which proves the claim.

Steps 2 and 3 (sketch; details in Section B). With $m = (n-1)/2$, $x = j - m$, $y = m - i$, the closed form gives $z = L(x, y) + n\,\zeta(x, y)$ with L affine and $\zeta = -n\lfloor (x - y + m)/n \rfloor - \lfloor (2x - y + m)/n \rfloor \in \{\pm(n+1), \pm 1, 0\}$. The affine bijection $\Phi(x, y, z) = (x, y, (z - L)/n)$ preserves the face lattice and the vertex cells, and $\Phi(\mathcal{G}(S))$ is the convex hull of the lattice points of five convex lattice polygons at the five heights ζ: two triangles T, $B = -T$ at $\zeta = \pm(n+1)$, two triangles U, $D = -U$ at $\zeta = \pm 1$, and a parallelogram M at $\zeta = 0$ (Figure 9a). Sixteen explicit planes, eight of them and their images under $(x, y, \zeta) \mapsto -(x, y, \zeta)$, are shown to support the convex hull of twelve named points $v_1, \ldots, v_{12}$—the three corners of T and of B, one corner each of U and D, and the four corners of

M—in facets that close up into a surface with $(V, E, F) = (12, 26, 16)$ closed under edge-adjacency, hence all facets (so $P = Q$), and every vertex of every layer polygon satisfies all sixteen inequalities (Figure 9b). These are finitely many polynomial identities and inequalities in m, verified exactly for all integers $m \geq 2$ by the script `siamese_proof.py`. Hence $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{v_1, \ldots, v_{12}\}$, and mapping back by Φ^{-1} gives the statement; for $m = 1$ the four triangles degenerate to points and the same argument gives the 8-vertex Lo Shu gem. □

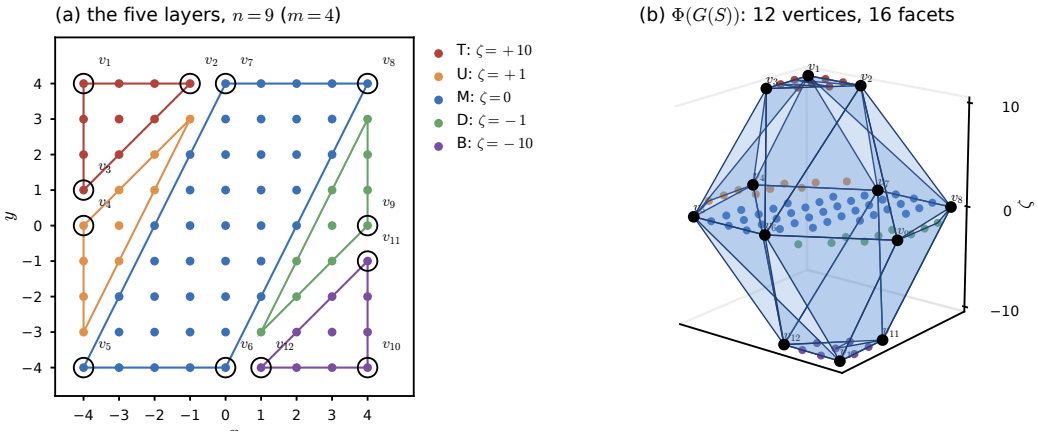


Figure 9: The Siamese gem of order $n = 9$ after the affine reduction of Proposition 5.3. (a) The 81 cells in the (x, y)-plane coloured by the layer value $\zeta \in \{+10, +1, 0, -1, -10\}$, with the five lattice polygons T, U, M, D, B outlined and the twelve vertex cells $v_1, \ldots, v_{12}$ circled. (b) The reduced polytope $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x, y, \zeta)\}$ with its 12 vertices and 16 facets (12 triangles, 4 quadrilaterals); the shapes of the layers do not depend on n except through scale, which is why the combinatorial type is constant for all odd $n \geq 5$.

6 Discussion and open questions

The Magic Gem is the convex hull of a magic square's height function over the centred grid, and the same polyhedron for all eight D_4-images of the square. The vector picture is exact but easily over-read: the centroid statement belongs to the full point set and holds for every arrangement; the magic conditions are the height-weighted ones; and the hull records only extent (Section 3). What the hull does see is symmetry: the complement map acts as a mirror reflection of the gem, and for order four this accounts for all the symmetry there is (Theorem 4.3) and for all coincidences between gems (Proposition 5.1). The coarse invariants, the vertex count (nine values) and the volume (155 values), are not well aligned with Dudeney's groups, and the vertex count is a property of the square, not of its group or construction (Dürer's square versus square #802).

Hull vertices for larger n (numerical). Table 4 lists the mean number of hull vertices of random arrangements (10^5 uniform samples for each $n \leq 10$, standard error at most 0.010; Qhull, with the vertex sets of the first 100 hulls of each batch re-checked in exact integer arithmetic), of the magic squares (exact for $n = 3, 4$; for $n = 5, \ldots, 8$ the mean over magic squares found by simulated annealing on the sum of squared line deviations, `anc/sa_magic.c`), and of the Siamese squares. The fraction $\mathbb{E}[V]/n^2$ of random arrangements falls from 0.91 ($n = 3$) to 0.28 ($n = 10$). At $n = 4$ magic gems have slightly more vertices than random ones (12.23 against 12.06 ± 0.004); for $n = 5, \ldots, 8$ the annealed means differ from the random means by at most 1.2 standard errors. This is a statement about the samples only: at $n = 4$, where all 880 classes are known, the annealer reached every class (20,000 draws, 10 to 41 hits per class, hit-weighted mean 12.24), but its sampling distribution for $n \geq 5$ is unknown and could be biased towards easily reached squares. The Siamese value 12 is thus atypical: it stays constant while the means grow (for $n = 7$ the smallest vertex count among 8000 annealed squares is 14, for $n = 8$ it is 16 among 2400).

Table 4: Mean number of hull vertices (numerical except where marked exact). Random: uniform random arrangements; magic: exact for $n = 3, 4$, annealing samples of 59,993, 24,000, 8,000, 2,400 squares for $n = 5, \ldots, 8$ (standard errors 0.007, 0.013, 0.025, 0.054); Siamese: exact.

n	3	4	5	6	7	8	9	10
random, $\mathbb{E}[V]$	8.22	12.06	15.39	18.39	21.08	23.53	25.79	27.91
magic	8 (exact)	12.23 (exact)	15.38	18.41	21.08	23.56	–	–
Siamese	8	–	12	–	12	–	12	–

Open questions.

1. *Order five.* By Schroeppel's enumeration there are 275,305,224 essentially different squares of order 5 [10]; an exact hull computation for each is feasible. How many combinatorial types occur, what is the modal f-vector, and is the complement map again the only source of congruence? By Proposition 4.2 the self-complementary squares of odd order are exactly the associative ones, so every symmetric gem of order five is centrally symmetric.
2. *Symmetry in general.* Is $\mathrm{Iso}(\mathcal{G}(S))$ generated by the complement symmetry for every magic square of every order, as it is for order four?
3. *Bounded hull-vertex counts.* Siamese gems have twelve vertices for every odd $n \geq 5$ because, affinely, they are hulls of five lattice polygons whose shapes do not depend on n (Proposition 5.3). Do the LUX, Strachey or diagonal-complement constructions have the same property, and is the fraction of vertices for a uniformly random magic square of large order bounded away from zero?
4. *Other square families.* The construction applies verbatim to semi-magic, pandiagonal and Franklin squares [13]; the statements of Proposition 3.2 already cover the semi-magic case. The gems of these families have not been examined.

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A Computational methods

Hardware and software. All computations were run on an Apple M4 Max (16 cores, 128 GB) under macOS, with Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1 (Quickhull only seeds the exact hull computation) and NetworkX 3.2.1. The scalar identities of Section 3 (the value $n^2(n^4-1)/12$ for the semi-magic squares and the product-of-inertia statistics over the 880 classes) were checked in exact integer arithmetic.

Exact hulls and their classification. Doubling all coordinates makes every point of a gem an integer vector. Facet planes are recovered from an initial Quickhull triangulation as primitive integer equations, coplanar triangles are merged into faces, vertices are the points on at least three facet planes, edges are vertex pairs sharing two facets, and volumes are exact rationals; Euler's relation $V - E + F = 2$ is checked for every gem. Combinatorial types are the isomorphism classes of the bipartite vertex–facet incidence graphs, grouped by a Weisfeiler–Lehman hash and confirmed by explicit isomorphism tests. Isometry groups (Theorem 4.3) are computed exactly as follows: the vertex set V (k points, doubled integer coordinates) is centred by $\mathbf{q} \mapsto k\mathbf{q} - \sum V$; an isometry of V is an orthogonal map fixing the origin, determined by the images of three linearly independent vertices, which are searched among the vertices with the same squared norm and mutual inner products; each candidate is accepted only if the induced bijection of V preserves all pairwise integer squared distances. Congruence is decided exactly by testing for a bijection of hull-vertex sets preserving all integer squared distances (grouping by the sorted distance multiset, then an edge-labelled graph isomorphism), with a floating-point Kabsch superposition as a cross-check. For the Siamese proof (Proposition 5.3, Section B) the finitely many polynomial identities and inequalities in m are verified by exact integer polynomial arithmetic, an inequality $p(m) > 0$ for all integers $m \geq 2$ being certified by positivity of the coefficients of $p(2+t)$ or, failing that, by exact evaluation up to a Cauchy root bound.

Reproducibility. The scripts for every table and figure and the JSON files containing all reported numbers are provided as ancillary files in the directory `anc/`, with a `README` listing the commands that regenerate them. Squares are numbered $0, \ldots, 879$ in the order of `anc/squares880.txt`. The web application at `https://kylemath.github.io/MagicGemWebpage/` displays the 880 essentially different order-four squares and renders the gem of a selected one.

B Siamese gems: the remaining steps of the proof

We continue the proof of Proposition 5.3 from the closed form $s(i,j) = n\langle i+j-m\rangle + \langle i+2j-2m\rangle + 1$.

Step 2: affine reduction to five layers. In gem coordinates $x = j - m$, $y = m - i$, with $\langle t\rangle = t - n\lfloor t/n\rfloor$,

$$z = \underbrace{n(x-y+m) + (2x-y+m) + 1 - \tfrac{n^2+1}{2}}_{L(x,y)\ \text{affine}} + n\,\zeta(x,y), \quad \zeta := -n\left\lfloor \tfrac{x-y+m}{n} \right\rfloor - \left\lfloor \tfrac{2x-y+m}{n} \right\rfloor.$$

The map $\Phi(x,y,z) = (x, y, (z - L(x,y))/n)$ is an affine bijection, so $\Phi(\mathcal{G}(S)) = \mathrm{conv}\{(x,y,\zeta)\}$ has the same face lattice and vertex cells as $\mathcal{G}(S)$. In the box $|x|, |y| \leq m$ both floors lie in $\{-1, 0, 1\}$, the pairs $(-1,1), (-1,0)$ and, by the central symmetry $\zeta(-x,-y) = -\zeta(x,y)$, $(1,-1), (1,0)$ are impossible, so ζ takes the five values $\pm(n+1), \pm 1, 0$ on the lattice points of five convex lattice polygons (Figure 9a): $T = \mathrm{conv}\{(-m,m), (-1,m), (-m,1)\}$ at $\zeta = n+1$ (cut out by $y \geq x+m+1$), $U = \mathrm{conv}\{(-m, 1-m), (-m,0), (-1, m-1)\}$ at $\zeta = 1$ (by $2x + m + 1 \leq y \leq x + m$), $M = \mathrm{conv}\{(-m,-m), (0,m), (m,m), (0,-m)\}$ at $\zeta = 0$ (by $|y - 2x| \leq m$), and $D = -U$, $B = -T$ at $\zeta = -1, -(n+1)$.

Step 3: the twelve points and sixteen planes. Let $v_1 = (-m, m, n+1)$, $v_2 = (-1, m, n+1)$, $v_3 = (-m, 1, n+1)$, $v_4 = (-m, 0, 1)$, $v_5 = (-m, -m, 0)$, $v_6 = (0, -m, 0)$, $v_7 = (0, m, 0)$, $v_8 = (m, m, 0)$, $v_9 = (m, 0, -1)$, $v_{10} = -v_1$, $v_{11} = -v_3$, $v_{12} = -v_2$ (the cells listed in the statement), and consider the inequalities

$$\begin{aligned}
&(1)\ \zeta \leq 2m+2, \qquad (2)\ y \leq m, \qquad (3)\ x \geq -m, \qquad (8)\ -2y + \zeta \leq 2m,\\
&(4)\ -2(m+1)x + (2m+1)y - m\zeta \leq m(2m+1),\\
&(5)\ 2(m+1)x - 2(m+1)y + (2m+1)\zeta \leq 2m(m+1),\\
&(6)\ 2m(m+2)x - (2m^2+2m-1)y + m(2m+1)\zeta \leq m(2m^2+2m-1),\\
&(7)\ 2mx - y + m\zeta \leq m(2m-1),
\end{aligned}$$

together with their images $(1')$–$(8')$ under $(x, y, \zeta) \mapsto -(x, y, \zeta)$; let Q be the polyhedron they define. The script `siamese_proof.py` verifies, as polynomial identities and inequalities in m valid for all integers $m \geq 2$ (Section A): (a) every vertex of T, U, M, D, B at its height satisfies all 16 inequalities, so $\Phi(\mathcal{G}(S)) \subseteq Q$; (b) each plane contains exactly the following v's, all others lying strictly inside:

$$(1)\colon v_1v_2v_3, \quad (2)\colon v_1v_2v_8v_7, \quad (3)\colon v_1v_3v_5v_4, \quad (4)\colon v_1v_4v_7,$$
$$(5)\colon v_2v_3v_6, \quad (6)\colon v_2v_6v_9, \quad (7)\colon v_2v_8v_9, \quad (8)\colon v_3v_5v_6,$$

with centrally symmetric lists for $(1')$–$(8')$; (c) the quadrilaterals are in convex position in the cyclic order given, so the 16 face polygons have 26 distinct edges, each shared by exactly two polygons; (d) each v_k lies on three planes with independent normals. By (b) each plane supports $P := \operatorname{conv}\{v_1, \ldots, v_{12}\}$ in a facet with the stated vertices; by (c) this family of facets is closed under edge-adjacency, and since the facet-adjacency graph of a 3-polytope is connected these are all the facets, so $P = Q$. With (a), $P \subseteq \Phi(\mathcal{G}(S)) \subseteq Q = P$, whose vertex set is $\{v_1, \ldots, v_{12}\}$ by (d); mapping back by Φ^{-1} gives the statement, with $(V, E, F) = (12, 26, 16)$ read off from (b)–(c). For $m = 1$ the triangles degenerate to points and the same argument gives the 8-vertex Lo Shu gem.